

Minimal Speed-Scale Orchestration (SSO)

Consciousness opens as cross-scale log referencing, not as a localized object.

Fast layer

high-frequency prediction / reaction / forward dynamics

V_f

dense updates

conscious access

boundary of cross-scale log referencing

$$\Delta v = v_f - v_s$$

finite speed difference

$$\tau \propto \Delta v / \mu$$

delay / buffering thickness

μ

coupling viscosity

only selected fast dynamics sediment into slow logs

neither complete synchrony nor complete separation

sparse stabilization

V_s

Slow layer

low-frequency memory / sedimentation / re-referenceable logs

$$0 < |\Delta v| < \Delta v_{\max}, \mu_{\min} < \mu < \mu_{\max}, \tau_{\min} < \tau < \tau_{\max}$$